

Mitchell Karmen

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Astrophysics PhD Candidate, **NSF GRFP Fellow**, Johns Hopkins University

Research interests: *Tidal disruption events at high redshifts, TDEs in LSST and Roman, SMBH seed and growth mechanisms, transient spectroscopic follow-up*

Education

PhD Expected 2028, Department of Physics and Astronomy, Johns Hopkins University.

Advisor: Suvi Gezari

M.S. 2025, Department of Physics and Astronomy, Johns Hopkins University.

B.A. 2020, Department of Physics, New York University. Advisors: David Hogg, Sjoert van Velzen, Glennys Farrar

Positions

Graduate Student, Johns Hopkins University, 2022–present.

Projects:

- COSMOS-Web Search for TDEs at High Redshifts
- SMURFS: Spectroscopic Monitoring of Ultraviolet Rapid Flares in Small galaxies
- SEDM-KP commissioning science team member

Adjunct Lecturer, Morgan State University, 2025.

Taught EASC101: Intro to Stellar Astronomy

Research Assistant, Lawrence Berkeley National Lab, 2021–2022.

SULI Intern, Lawrence Berkeley National Lab, 2020–2021

Software Developer, Zwicky Transient Facility, 2020.

Observing Programs as PI

James Webb Space Telescope Cycle 3 DDT: 9.5 hours

2025 PI: DD 9356, HZTDE-1: The Highest-Redshift TDE Candidate, Revealing the Origins of Supermassive Black Holes

Apache Point Observatory: 11 hours

2025 PI, Spectroscopic Monitoring of Ultraviolet Rapid Flares in Small galaxies

Publications

First-author publications

- Karmen, Mitchell**; Gezari, Suvi; Norman, Colin; & Guolo, Muryel, 2026, *Tidal disruption event rates across cosmic time: forecasts for LSST, Roman, and JWST and their constraints on the supermassive black hole mass function*, ArXiv (arXiv:2602.04947)
- Karmen, Mitchell**; Gezari, Suvi; Lambrides, Erini; Akins, Hollis B.; *et al.*, 2025, *JWST Discovery of a High-redshift Tidal Disruption Event Candidate in COSMOS-Web*, The Astrophysical Journal, **990**, 149 (arXiv:2504.13248) [11 citations]
- Karmen, Mitchell**; Gezari, S.; Gomez, S.; Guolo, M; Norman, C; 2023, *Roman CCS White Paper: A Sweet Spot for Tidal Disruption Events with the Roman Space Telescope*

Co-authored publications

- 14 Guolo, M.; Mummery, A.; van Velzen, S.; Gezari, S.; *et al.* (incl. **MK**), 2025, *Compact Accretion Disks in the Aftermath of Tidal Disruption Events: Parameter Inference from Joint X-ray Spectra and UV/Optical Photometry Fitting*, ArXiv (arXiv:2510.26774) [6 citations]
- 13 Zhang, Yiyang; Ding, Xuheng; Yang, Lilan; Lambrides, Erini; *et al.* (incl. **MK**), 2025, *Unveiling Extended Components of 'Little Red Dots' in Rest-Frame Optical*, ArXiv (arXiv:2510.25830) [5 citations]
- 12 DeCoursey, Christa; Egami, Eiichi; Sun, Fengwu; Akhtarkavan, Arshia; *et al.* (incl. **MK**), 2025, *The First Photometric Evidence of a Transient/Variable Source at $z \gtrsim 5$ with JWST*, The Astrophysical Journal, **990**, 31 (arXiv:2504.17007) [9 citations]
- 11 Franz, Noah; Alexander, Kate D; Gomez, Sebastian; Christy, Collin T; *et al.* (incl. **MK**), 2025, *The Open multiwavelength Transient Event Repository (OTTER): Infrastructure Release and Tidal Disruption Event Catalog*, ArXiv (arXiv:2509.05405) [5 citations]
- 10 Hand, Jared; Kim, A. G.; Aldering, G.; Antilogus, P.; *et al.* (incl. **MK**), 2025, *An Agnostic Approach to Building Empirical Type Ia Supernova Light Curves: Evidence for Intrinsic Chromatic Flux Variation Using Nearby Supernova Factory Data*, The Astrophysical Journal, **982**, 110 (arXiv:2505.07880)
- 9 Pierel, J. D. R.; Coulter, D. A.; Siebert, M. R.; Akins, H. B.; *et al.* (incl. **MK**), 2025, *Testing for Intrinsic Type Ia Supernova Luminosity Evolution at $z \gtrsim 2$ with JWST*, The Astrophysical Journal, **981** (arXiv:2411.11953) [15 citations]
- 8 Akins, Hollis B.; Casey, Caitlin M.; Berg, Danielle A.; Chisholm, John; *et al.* (incl. **MK**), 2025, *Strong Rest-UV Emission Lines in a "Little Red Dot" Active Galactic Nucleus at $z = 7$: Early Supermassive Black Hole Growth alongside Compact Massive Star Formation?*, The Astrophysical Journal, **980** (arXiv:2410.00949) [51 citations]
- 7 Coulter, D. A.; Pierel, J. D. R.; DeCoursey, C.; Moriya, T. J.; *et al.* (incl. **MK**), 2025, *Discovery of a likely Type II SN at $z \approx 3.6$ with JWST*, ArXiv (arXiv:2501.05513) [12 citations]
- 6 Gomez, Sebastian; Nicholl, Matt; Berger, Edo; Blanchard, Peter K.; *et al.* (incl. **MK**), 2024, *The Type I superluminous supernova catalogue I: light-curve properties, models, and catalogue description*, Monthly Notices of the Royal Astronomical Society, **535**, 471 (arXiv:2407.07946) [36 citations]
- 5 Siebert, M. R.; DeCoursey, C.; Coulter, D. A.; Engesser, M.; *et al.* (incl. **MK**), 2024, *Discovery of a Relativistic Stripped-envelope Type Ic-BL Supernova at $z = 2.83$ with JWST*, The Astrophysical Journal, **972** (arXiv:2406.05076) [16 citations]
- 4 Pierel, J. D. R.; Engesser, M.; Coulter, D. A.; DeCoursey, C.; *et al.* (incl. **MK**), 2024, *Discovery of an Apparent Red, High-velocity Type Ia Supernova at $z = 2.9$ with JWST*, The Astrophysical Journal, **971** (arXiv:2406.05089) [27 citations]
- 3 Gomez, Sebastian; Alexander, Kate; Berger, Edo; Blanchard, Peter K.; *et al.* (incl. **MK**), 2023, *Roman CCS White Paper: Characterizing Superluminous Supernovae with Roman*, ArXiv (arXiv:2306.17233) [2 citations]
- 2 Aldoroty, L.; Wang, L.; Hoefflich, P.; Yang, J.; *et al.* (incl. **MK**), 2023, *Bump Morphology of the CMAGIC Diagram*, The Astrophysical Journal, **948**, 10 (arXiv:2210.06708) [3 citations]
- 1 Stein, George; Seljak, Uroš; Böhm, Vanessa; Aldering, G.; *et al.* (incl. **MK**), 2022, *A Probabilistic Autoencoder for Type Ia Supernova Spectral Time Series*, The Astrophysical Journal, **935**, 5 (arXiv:2207.07645) [25 citations]

Talks & presentations

A candidate high- z tidal disruption event at $z \approx 5$ in COSMOS-Web: the first photometric search with JWST,

2025, Contributed Talk, Massive Black Holes across Cosmic Time, Cambridge, UK

A Candidate High-redshift Tidal Disruption Event in the JWST COSMOS-Web Survey,

2025, Invited Talk, High Energy Astrophysical Transients Seminar, University of Maryland College Park, College Park, MD

Searching for Tidal Disruption Events at High Redshift with the JWST COSMOS-Web Survey,

2025, Contributed Talk, AAS 245, National Harbor, MD

High Redshift TDEs: Probes of Low Mass SMBHs at High Redshifts,

2024, Poster, Joint Space-Science Institute Workshop, Baltimore, MD

A new window into the first black holes with JWST observations of high-redshift TDEs,

2024, Contributed Talk, COSMOS-Web Collaboration Meeting, IPMU, Tokyo, Japan.

2024, Contributed Talk, First Stars VII, Center for Computational Astrophysics, New York City.

Enabling Discovery of High Redshift TDEs through Roman,

2023, Invited Talk, Roman Community Forum, Virtual.

Extending the Twins Embedding for Use by the Roman/WFIRST Supernova Survey,

2021, Contributed Talk, Roman Science Team Community Briefing, Virtual.

Investigating the Variability of Type Ia Supernovae Using Deep Learning,

2021, Poster, Lawrence Berkeley Lab Summer Intern Poster Session, Berkeley, CA.

A New All-Sky Catalog of Compact Galaxies,

2020, Poster, American Astronomical Society Meeting 225, Honolulu, HI.

A New All-Sky Catalog of Compact Galaxies: Opportunities for Electromagnetic Transient Follow-Up,

2019, Poster, Gotham AstroFest, New York, NY.

Mentoring

Jeevika Setzer- JHU undergraduate student (Fall 2024–Present)

Seti Norris- STScI SASP Intern (Summer 2024)

Awards and Honors

NSF Graduate Research Fellowship, 2024.

George Granger Brown Award (NYU), 2020.

Dean's Undergraduate Research Award (NYU), 2019.

Teaching

Adjunct Lecturer, Morgan State University, 2025.

EASC101: Intro to Stellar Astronomy

Physics and Astronomy Teaching Assistant

General Physics Lab II, Johns Hopkins University, 2023.

Planets, Life and the Universe, Johns Hopkins University, 2022.

General Physics Lab I, Johns Hopkins University, 2022.
Berkeley Lab Director's Apprenticeship Program, Volunteer Coding Instructor, 2021,
Department Tutor (NYU)
General Physics I: Kinematics General Physics II: Electricity and Magnetism
7th Grade Mathematics Teaching Assistant, America Counts, 2017–2018.

Outreach and Leadership

Black (hole) Coffee, Johns Hopkins University, 2024–Present
Lead Organizer
Committee for Diversity & Inclusion, Johns Hopkins University 2024–Present,
Committee Member
Johns Hopkins University / Space Telescope Science Institute Liason 2023–2024,
I organized joint events between graduate students at JHU and scientists at STScI
Berkeley Lab Director's Apprenticeship Program, 2021,
Helped to instruct low-income high school students in the basics of Python programming;
mentored a small group of students in applying to college, writing application essays, etc.
Tampa Heights Junior Civic Association, 2021,
One-on-one mentor to a low-income middle school student who is interested in science.
NYU Society of Physics Students 2019–2020,
President
NYC BioBus 2019–2020,
Volunteer Instructor; Led groups of high school students through physics demonstrations

Additional Education

Flatiron LSST Summer School: The landscape of astronomical transients, 2025.
Summer School in Statistics for Astronomers XVII, Penn State University, 2023.
Center for Teaching Excellence and Innovation Teaching Institute, Johns Hopkins University,
2023.
GROWN Multi-Wavelength Astronomy Summer School, Virtual, 2020.
Data Analysis Boot Camp, Center for Computational Astrophysics, 2019.